

RAMS — User Guide & Handbook

RAMS (Road Asset Management System) is an Indian pavement deterioration, remaining-life and investment-planning engine.

RAMS (Road Asset Management System) is an Indian pavement deterioration, remaining-life and investment-planning engine. It takes you from pavement design -> field-condition data -> deterioration forecasting -> investment decisions (life-cycle analysis, network budgeting, PBMC pricing), with every rule citing its IRC clause and every cost grounded in the MoRTH Standard Data Book.

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This guide explains **how to use each feature, what data it needs, the accepted file formats, and how the calculations are done.**

1. Getting started

RAMS is pure-Python (standard library only — no installation of third-party packages).

Run the dashboard

```
python -m rams.server --port 8000
# then open http://127.0.0.1:8000
```

Command line (batch / automation)

```
python -m rams.cli --design --cbr 8 --design-msa 120 --design-life 15
python -m rams.cli --fwd examples/fwd_sections_sample.csv --design-msa 300
python -m rams.cli --pbmc --term-years 5          # see --help for all flags
```

The dashboard has **four tabs**: **Segment Forecast**, **Network & Budget**, **Calibrate & Residual Life**, and **Design & PBMC**.

2. Data formats (general rules)

These rules apply everywhere a file is accepted:

Format	Notes
CSV	A header row plus data rows. Most universal.
XLSX	Multiple sheets are read; merged/grouped two-row headers are combined; the <i>*first non-empty*</i> value wins for duplicate columns.
PDF	Text-based (machine-readable) PDFs only — the table is recovered from the text layer. Scanned/image PDFs are not OCR'd.

Column naming is forgiving. Headers are normalised (lower-cased, spaces and newlines become _), and each feature keeps a list of **aliases** so common template variations are recognised automatically.

Missing optional columns fall back to documented defaults rather than erroring — only the **primary** field(s) of each feature are required.

One upload, many tabs. A survey loaded in **Segment Forecast -> Load a segment** is shared with the Network, PBMC and LCA tabs (look for the green **“Use uploaded survey”** button), so you don't re-upload.

3. Tab: Segment Forecast

Purpose: project a single segment (or a loaded survey) forward in time and see when it crosses maintenance thresholds.

Inputs

- Condition: IRI (mm/m), rut (mm), cracking (% area), and traffic (annual MSA, growth). Monsoon zone (HIGH/MEDIUM/LOW) scales environmental deterioration.
- Optional structural inputs (FWD deflection, structural number) when using the HDM-4 rut model.

Load a segment from a survey / FWD file

Upload one or more **.csv / .xlsx / .pdf** files. RAMS:

- detects rut / roughness / cracking / pothole surveys and ****merges them by chainage**** into one dataset;
- builds **homogeneous sections** and shows a paginated table;
- fills traffic/zone with defaults if the survey doesn't carry them (it usually doesn't — an NSV survey is condition-only). A provenance note states exactly which columns came from the file vs. defaults.

What you get

- **PCI curve** (IRC:82) with decision-band shading.
- **Untreated vs managed** trajectories (the managed line applies treatments and resets condition).
- **Intervention triggers**, HDM-4 breakdowns, skid, potholes, and an MCI cross-reference — each in a paginated table.
- **Treatments applied** with **MoRTH-based costs**.
- Download the homogeneous-section report as **XLSX / PDF**.

How it's calculated

- **PCI** from IRI, rut and cracking per IRC:82.
- **Rut / cracking / roughness** progress year-by-year using IRC:82 power-law defaults or **HDM-4** models (driven by traffic and, under HDM-4, FWD deflection / structural number).
- **Triggers** fire at IRC thresholds (rut, cracking, IRI, deflection, MSA).

4. Tab: Network & Budget

Purpose: allocate a fixed annual budget across many segments to maximise avoided structural cost.

Inputs

- A network: the demo network, an imported file, or the **uploaded survey**.
- Annual budget (Rs lakh), unit cost, horizon, rut model, design MSA, and an optional handback requirement (residual MSA).

Import / preview

Upload **.csv** / **.xlsx** / **.pdf**. The preview table is paginated; a note shows which columns are file-derived vs defaulted.

What you get

- KPIs: total spend, avoided structural cost, segments funded / **unfunded**.
- **Spend-by-year** bars vs the annual budget.
- A paginated **treatment schedule** and a **network risk / residual-life** table.
- **“Unfunded”explainer**: the recommended annual budget that would clear the backlog (so the warning becomes an actionable number).

How it's calculated

- Each segment is forecast; those entering their preventive window become candidates, ranked by exposure (MSA × length), deadline, then cost.
- A greedy multi-year allocation funds each candidate in the first in-window year with budget left; the rest are **unfunded** (they fall to costlier structural repair later).
- The recommended budget is found by binary-searching for the smallest annual budget that leaves zero unfunded.
- **Residual life** = the governing minimum of IRC:81 deflection capacity and the IRC:37 traffic budget.

5. Tab: Calibrate & Residual Life

5a. Calibrate a deterioration model

Purpose: fit a deterioration model's coefficients to **your** field data by OLS regression, so forecasts match local behaviour.

Pick the model, then paste or upload observations (.csv / .xlsx / .pdf).

Model	Required columns	Optional (defaulted)
Rutting (Krid/Krst/Krpd)	`measured_rut_increment_mm` `ye4`, `age`	`deflection_mm`, `structural_number`, `compaction_pct`, `cds`, `heavy_speed_kmh`, `surfacing_thickness_mm`
Cracking (MLIT a,b)	`crack_prev`, `crack_next`	—
Roughness (HDM-4)	`measured_iri_increment`, `iri`, `age`	`structural_number`, `d_msa`, `d_crack_pct`, `d_rut_mm`
Skid (decay_k)	`measured_sfc_decrement`, `sfc`	`d_msa`
Potholes (rate)	`measured_pothole_increment`, `cracking_pct`	`d_msa`

- **Column aliases** are recognised (e.g. Rut Increment mm, ESA MSA->ye4, SNP->structural_number, crack_t0/crack_t1).

- Each model only consumes rows (from any sheet) that carry its target column; bad rows are skipped, not fatal.

Important — calibration needs *change over time.* Calibration fits a deterioration **rate**, so it requires either repeat surveys of the same segment (year-on-year increments) or per-segment age + traffic. A single NSV condition snapshot only has current condition — RAMS detects this and tells you to use that file in the Forecast / Sections / Network / LCA tabs instead.

How it's calculated: ordinary least squares on the model's linear form, with a non-negativity refit (coefficients that come out negative are pinned to zero and the rest re-fit). Reports the coefficients, R^2 , RMSE before/after, and rows used.

5b. Residual life

Enter FWD deflection, traffic and (optionally) design MSA and a required residual MSA. Returns the **IRC:81 deflection capacity vs IRC:37 traffic budget**, the governing minimum, and a **HAM handback verdict** (PASS / MARGINAL / FAIL) with the overlay deflection target needed to pass.

6. Tab: Design & PBMC

6.1 IRC:37 pavement design (CBR -> layer thicknesses)

Inputs: subgrade CBR, design traffic (design MSA directly, or $CVPD \times VDF \rightarrow MSA$ via IRC:37), design life, carriageway, and method (**Catalogue** or **IITPAVE mechanistic**).

Outputs: BC / DBM / WMM / GSB thicknesses, subgrade modulus, a **to-scale cross-section diagram**, and (mechanistic method) a fatigue/rutting strain check.

How it's calculated:

- ***Catalogue*** — the IRC:37-2018 design catalogue section for the CBR and traffic.
- ***Mechanistic*** — sizes the **least-cost** section whose Odemark–Boussinesq fatigue (ϵ_{ps_t} at the bottom of bituminous) and rutting (ϵ_{ps_v} at the subgrade top) strains both satisfy the design traffic, calibrated to IITPAVE.

6.2 IITPAVE section check

Enter layer moduli and thicknesses -> tensile/vertical strains and fatigue/rutting life (IRC:37 or IRC:115 standard). Used to confirm a design or a back-calculated section.

6.3 FWD remaining-life & overlay

Purpose: from an FWD report's homogeneous sub-sections, compute remaining life and flag overlays.

- Paste the sections, or upload the report (**.csv / .xlsx / .pdf**).
- **Columns:** section_id, e_bituminous (**required**), e_granular, e_subgrade, h_bituminous, h_granular (optional — **default if absent**), plus optional chainage_from/to. Aliases like E_BC, E_base, E_sg, H_BC, H_base are recognised.
- **How it's calculated:** Odemark–Boussinesq strains -> IRC:115-2014 remaining fatigue/rutting life; sections below the design traffic need an overlay, and those within 15% of the threshold are marked ***confirm with IITPAVE*** (screening-grade, $\sim \pm 10\%$ of full IITPAVE).

6.4 PBMC estimate (5–7 yr)

Prices a Performance-Based Maintenance Contract to hold a service level. **Inputs:** term, performance PCI, routine rate, escalation, contingency, overhead, discount rate (and a segment or the uploaded survey).

Outputs: per-year cash flow (routine + periodic + initial rectification), contract value, cost per km, NPV, and a service-level compliance check.

6.5 Life-Cycle Analysis (LCA) & decision matrix

Purpose: project one segment over a horizon and decide, each year, the treatment that keeps it serviceable — at MoRTH cost.

- **Decisions:** ROUTINE -> PREVENTIVE -> OVERLAY -> RECONSTRUCTION, triggered by PCI / rut / cracking / roughness thresholds, with condition reset after a major treatment and a minimum-interval deferral.
- **Costs:** MoRTH SDB rate × carriageway area; reports total, **NPV** and **EUAC** (equivalent uniform annual cost).
- Download the matrix as **XLSX / PDF**.

7. Costing basis (MoRTH Standard Data Book)

All maintenance/treatment costs use **indicative MoRTH SDB unit rates** (Rs /m²): routine Rs 60, microsurfacing Rs 160, mill & overlay Rs 820, reconstruction Rs 3200, each × carriageway area (length × width, default 7 m). These rates and all thresholds are **editable data, not hard-coded logic** — replace them with the current SDB / State Schedule of Rates before tendering.

8. Outputs & reporting

- Interactive tables are **paginated** (25 / 50 / 100 / all rows) so large surveys load fast.
- Homogeneous sections, LCA matrices and forecasts export to **XLSX and PDF**.
- Reports are **self-contained and offline** — they open on an air-gapped workstation and can be emailed as a single file.

9. Limitations & honest caveats

- **Mechanistic strains are screening-grade** (Odemark–Boussinesq). Confirm borderline sections with full IITPAVE before construction/tender.
- **MoRTH rates are indicative.** Load the current SDB / State SoR for tendering.
- **Calibration needs repeat surveys** (or age + traffic). A single condition snapshot feeds forecasting, not rate-fitting.
- **PDF ingestion** requires a text layer; scanned images are not OCR'd.
- Defaults (traffic MSA, monsoon zone, lane width, layer thicknesses) are clearly flagged where used; override them with real values for project-grade results.

10. Quick reference — required columns

Feature	Required	Common optional (defaulted)
Survey / Forecast / Sections	chainage + at least one of rut / IRI / cracking	traffic MSA, monsoon zone, growth, length
FWD overlay	`e_bituminous`	`e_granular`, `e_subgrade`, `h_bituminous`, `h_granular`, chainage
Calibrate (rut)	`measured_rut_increment_mm`, `ye4`, `age`	deflection, SNP, compaction, cds, speed, surfacing
Calibrate (cracking)	`crack_prev`, `crack_next`	—
Calibrate (roughness)	`measured_iri_increment`, `iri`, `age`	SNP, d_msa, d_crack_pct, d_rut_mm
Network & Budget	a network (file or uploaded survey)	budget, horizon, design MSA
Design	CBR + design traffic	VDF, design life, carriageway